RadOncReason: Reinforcement Learning with Verifiable Rewards for Prostate Cancer Treatment Planning

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Background

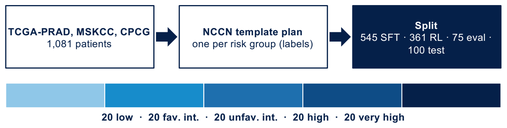
- Large language model (LLM) treatment recommendations are fluent but not always correct (Chen 2023).
- Using NCCN (National Comprehensive Cancer Network) guidelines as an example, can verifiable rewards teach LLMs to reason?

Inputs
PSA
Grade Group
T-stage
Biopsy cores
PSA density

Very low cT1c, GG1, PSA <10, <3 cores, PSA density <0.15	Active surveillance; EBRT or brachytherapy; RP
Low cT1–cT2a, GG1, PSA <10	Active surveillance; EBRT or brachytherapy; RP
Intermediate cT2b–c, GG2–3, or PSA 10–20. Favorable: 1 factor, GG1–2. Unfavorable: 2–3 factors, GG3 or ≥50% cores	**Favorable:** surveillance; RP; EBRT or brachytherapy **Unfavorable:** RP + PLND; EBRT + ADT 4–6 mo; EBRT + brachytherapy boost
High cT3a, GG4–5, or PSA >20	EBRT + ADT 1.5–3 y; EBRT + brachytherapy boost + ADT; RP + PLND
Very high cT3b–T4, primary pattern 5, ≥2 high-risk features	EBRT + ADT 1.5–3 y ± abiraterone; EBRT + brachytherapy boost + ADT; RP + PLND

NCCN prostate risk strata (the verifier's rule set): GG = Grade Group; EBRT = external-beam radiotherapy; RP = prostatectomy; PLND = node dissection; ADT = androgen deprivation.

Methods



Data: Test set balanced, 20 per risk group; ten-site extension 5,267 patients, 523 held out.

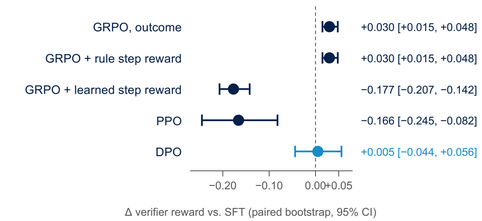
Verifier

- Each plan field (modality, dose, ADT) is scored against the patient's NCCN stratum:

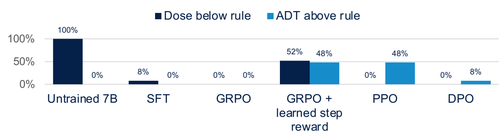
Verifier outcome, per plan field	Reward
Matches an option for the patient's stratum	**1.0**
Valid plan, but for a different stratum	**0.3**
Outside the rule set	**0.0**

Results

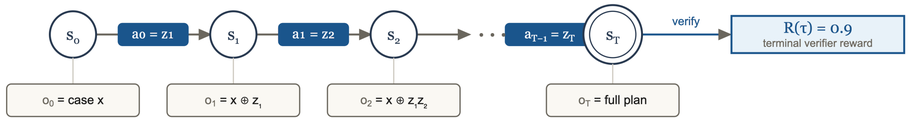
- A verifiable, rule-based reward reaches full verifier reward; learned proxy rewards and PPO drift outside the rule set.



Change vs. SFT: Navy = CI excludes 0. Step rewards also score each reasoning step (keyword rules or a trained model).

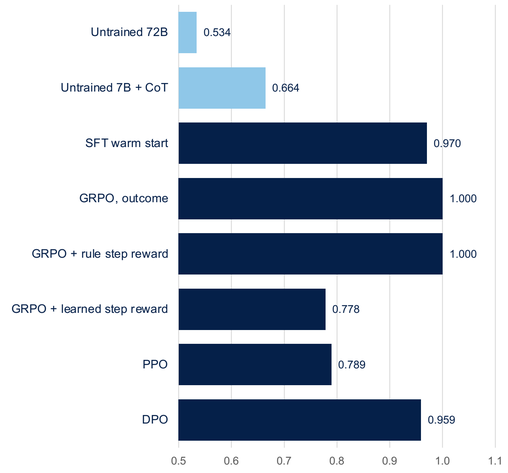


Verifier failures: Untrained models fall below the rule's dose; learned proxies exceed its ADT duration.

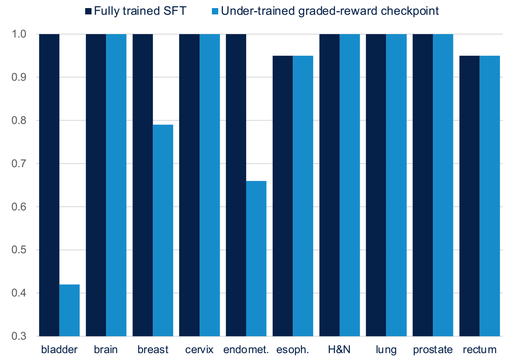


Token-level MDP (Markov decision process): Policy: Qwen2.5-7B. State = prompt plus tokens so far; action = next token; one terminal reward over plan fields f (modality, dose, ADT): R(τ) = 0.9 Σf wf cf(τ) + 0.1 R format.

Verifier reward



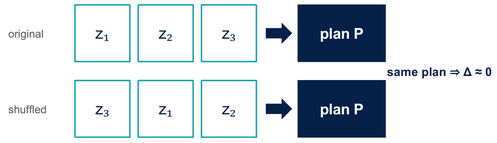
Prostate, by method: Only the verifiable reward improves on SFT; scale alone does not. CoT = chain of thought; mean of 3 seeds.



Ten disease sites (N = 5,267): SFT is near-perfect everywhere; an under-trained checkpoint scores lower at bladder, endometrium and breast.

Key findings

- GRPO with the verifiable reward makes no verifier errors, so the verifier works as an RL signal.
- GRPO improves on a strong SFT warm start, which a 10× larger untrained model does not.
- Learned proxy rewards invite hacking, pushing ADT outside the rule set on nearly half of cases (Gao 2023; Skalse 2022).
- Shuffling the stated rationale leaves plans unchanged, so the reasoning is not faithful (Turpin 2023; Lanham 2023).



Faithfulness probe: Shuffle the reasoning, re-decide, measure change.

Conclusions

- An executable rule-based verifier gives RL a checkable reward for multi-step plans across ten disease sites.
- Verifiable rewards help; learned proxies invite over-optimization.
- Next steps: learn the distribution of actions of clinicians.

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